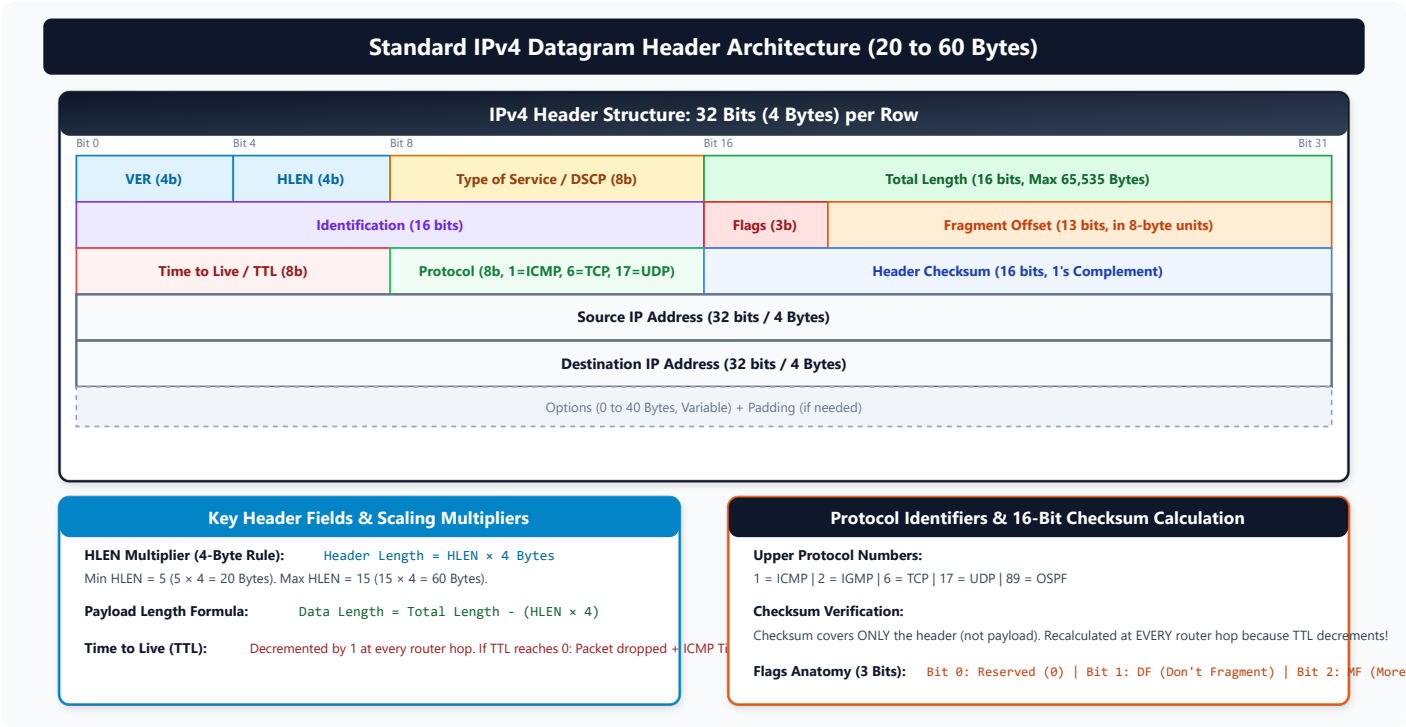


Module 02: IPv4 Header Format & Protocol Architecture

Unit 3: Network Layer

IPv4 Datagram Specification

Field-by-field architectural decomposition of the standard 20 to 60-byte IPv4 header, scaling factors, TTL loop protection, protocol multiplexing, and solved packet dissection numericals.



1. The 32-Bit Word Header Structure

IPv4 organizes header fields into 32-bit (4-byte) rows. HLEN measures length in 4-byte increments (5 to 15), dictating a header size between 20 and 60 bytes. Total Length spans up to 65,535 bytes.

2. Fragmentation and Lifetime Fields

Fundamental Equations

Header Size (Bytes) = $HLEN \times 4$

Payload Size = Total Length - $(HLEN \times 4)$

Byte Offset = Fragment Offset $\times 8$

Hop Cost: $TTL_new = TTL_old - 1$ (Dropped when $TTL == 0$)